Injecting Label Structure into Embedding Space for Ordinal and Circular Speech Classification

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Abstract. There are predictive machine learning classification tasks in which the predicted outputs exhibit a particular structure. For example, age detection and speech intelligibility prediction can be formulated as ordinal classification problems, and speech emotion recognition can be formulated as a circular classification problem. Existing classification losses typically treat classes as independent categories and do not explicitly account for these relationships. In this paper, we generalize margin-based softmax losses by introducing a class topology matrix that can encode complex class relationships including ordinal and circular topologies to learn a structurally consistent embedding space. In structured tasks where misclassification costs depend on the distance between classes, learning class anchors that reflect the underlying label topology improves performance while reducing large-distance misclassifications. Our experiments on three speech tasks demonstrate that the proposed loss functions achieve performance comparable to CORAL and CORN on standard ordinal classification, and outperform classification baselines on a circular classification problem.

Keywords: Angular margin · Ordinal classification · Circular classification · Age detection · Lyrics intelligibility · Speech emotion recognition · Metric learning.

1 Introduction

Predictive machine learning tasks are typically formulated as either classification in which the output space is discrete, or as regression in which the output space is continuous. For classification tasks, it is often assumed that any additional topological structure in the labels will be implicitly captured by the model without having to enforce it explicitly. Nevertheless, methods using prior knowledge of a known label structure have been shown to improve model performance. For example, approaches specifically designed for ordinal regression and classification have been proposed [25, 16, 26] and models such as CORAL [5] and CORN [35] outperform baseline approaches that do not take into account the output class ordering. The output class ordering is also reflected on different types of prediction errors. For example, misclassifying age 65+ as <20 vs. as 40-65 can have very different practical implications.

In this paper, we show how loss functions based on angular margin can be generalized through the use of a class topology matrix to model different output class topologies. We focus on two ordinal regression speech-related tasks and on one circular classification task, in which the labels are assumed to be arranged on a circle with periodic adjacency, so that the first and last classes are neighbours. The proposed loss functions achieve performance comparable to current ordinal algorithms and outperform alternatives on the circular classification task.

2 Related Work

Methods developed for problems with ordinal label structure are shown to outperform structure-agnostic methods. Therefore, a common strategy for continuous prediction is to reformulate it as an ordinal regression or ordered classification problem. For example, Niu et al. [26] modeled age prediction as a set of ordered binary classification problems using a multi-output CNN. Diaz et al. [11] showed that soft-label encoding of ordered targets can further improve ordinal modeling performance. Cao et al. [5] introduced a rank-consistent ordinal regression framework (COnsistent RAnk Logits - CORAL) which formulates ordinal regression as a set of $K-1$ binary classifiers with shared logits, extracting the monotonicity of predictions across class orders. Finally, Shi et al. [35] proposed a Conditional Ordinal Regression for Neural Networks (CORN) that trains ordinal regression through conditional binary problems, modeling the probability of exceeding each class threshold and ensuring rank consistency of predictions.

However, the class relationship structure is often unknown or ignored in classification tasks. Softmax loss is used across multiple fields to learn separable features for classification problems, but does not perform as well when inter-class similarity and intra-class variation are high, as in the case of speaker recognition [44, 27] and speaker verification [45, 15]. In these cases, it is desirable to satisfy the constraint that the largest intra-class distance is strictly less than the smallest inter-class distance with respect to a specified metric [18]. To alleviate this problem, the softmax loss has been combined with Euclidean margin-based losses [37, 38, 43]. However, Euclidean margin assumes isotropic Gaussian clusters, while softmax produces directional separation as the features learned have an innate angular distribution, as observed in [18] and verified in [43].

Following this observation, the Angular Softmax loss (A-Softmax) [19, 18] proposed a multiplicative angular margin penalty to simultaneously enforce intra-class compactness and inter-class dissimilarities. The authors note that the decision boundary in the softmax loss of a binary-class case is $(\boldsymbol{W}_1 - \boldsymbol{W}_2)\boldsymbol{x} + b_1 - b_2 = 0$ where $\boldsymbol{W}_i$ and b_i are the weights and biases of the softmax loss, with the constraint $||\boldsymbol{W}_1|| = ||\boldsymbol{W}_2|| = 1$ and $b_1 = b_2 = 0$, the decision boundary becomes $||x||(\cos(\theta_1) - \cos(\theta_2)) = 0$ where θ_i is the angle between $\boldsymbol{W}_i$ and $\boldsymbol{x}$. By introducing the multiplicative margin m, the decision boundaries become $||x||(\cos(m\theta_1) - \cos(\theta_2)) = 0$ and $||x||(\cos(\theta_1) - \cos(m\theta_2)) = 0$ for class 1 and 2, respectively. However, the non-monotonicity of the cosine function makes the optimization difficult, and different margins are created for different classes.

Additive Margin Softmax (AM-Softmax) [41] was introduced as a simpler, more interpretable and better-performing way to incorporate the angular margin into the softmax loss. After normalizing both the feature and the weight, the input becomes $x = \cos(\theta_{y_i}) = \frac{W_{y_i}^T f_i}{\|W_{yi}\|\|f_i\|}$, the cosine similarity between weight W_{y_i} and the feature vector f_i. Therefore, the predictions depend only on the angle θ_{y_i} and not on magnitude. The penalty m is applied to the cosine similarity. A scaling factor s is introduced following recommendations found in [42, 20, 21].

Additive Angular Margin Loss (AAM-Softmax) [10] is based on the observation that the predictions depend only on the angle θ_{y_i} following the normalization steps on both the feature and the weight. Therefore, the learned embedding features lie on a hypersphere of radius s. The authors add the penalty m directly between the weight W_{y_i} and the feature vector f_i corresponding to the geodesic distance margin penalty in the normalized hypersphere.

3 Proposed Approach

The proposed methods are built upon the strategies employed in AM-Softmax [41] and AAM-Softmax [10] losses to reduce intra-class variation by penalizing the target class margin. To encode label topology in the class anchors, we introduce additional penalty terms on non-target logits that encourage a structurally consistent embedding space. The canonical softmax loss function for classification is defined as:

$$\mathcal{L}_{\mathrm{S}} = -\frac{1}{n}\sum_{i=1}^{n}\log\frac{e^{W_{y_i}^T \boldsymbol{f}_i}}{\sum_{j=1}^{C} e^{W_j^T \boldsymbol{f}_i}} \tag{1}$$

where n is the number of training samples, C is the number of classes, $f_i = x_i + b_{y_i}$ is the input of the last fully connected layer, W_j denotes the j-th column of the last fully connected layer, W_{y_i} the column corresponding to the true class of y_i. Many authors refer to $W_{y_i}^T \boldsymbol{f}_i$ as the target logit of the i-th sample [29, 41, 10]. AM-Softmax and AAM-Softmax both use a margin penalty to reduce the intra-class variation. They can be defined as:

$$\mathcal{L}_{\mathrm{A(A)M}} = -\frac{1}{n}\sum_{i=1}^{n}\log\frac{e^{\,s\psi}}{e^{\,s\psi} + \sum_{\substack{j=1 \\ j\neq y_i}}^{C} e^{\,s\Psi}} \tag{2}$$

where

$$\psi = \begin{cases} \cos(\theta_{y_i}) - m & \text{if } \mathcal{L}_{\mathrm{AM}} \\ \cos(\theta_{y_i} + m) & \text{if } \mathcal{L}_{\mathrm{AAM}}, \end{cases} \tag{3}$$

$$\Psi = \cos(\theta_j), \tag{4}$$

m is a penalty, C is the number of classes, s is a scaling factor representing the radius of the projection and

$$\cos(\theta_{yi}) = \frac{W_{yi}^T f_i}{\|W_{yi}\|\,\|f_i\|} \tag{5}$$

is the cosine similarity between the feature vector f_i and the true class y_i represented by weight vector W_{y_i}. Both use a margin penalty on the true class, but differ in where they add it. While AM-Softmax applies the penalty m on the cosine similarity $\cos(\theta_{y_i})$, AAM-Softmax adds its margin penalty to the angle θ_{y_i} directly, enforcing the penalty between f_i and W_{yi}.

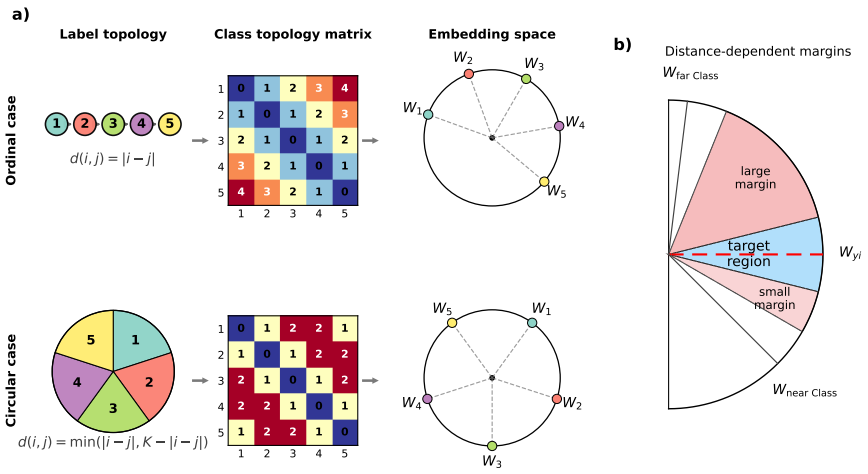


Fig. 1. How SIA(A)M loss functions shape the embedding space. a) Ordinal (top) and circular (bottom) label topologies define a Class Topology Matrix (CTM) that structures the embedding space and class anchors in the model's final layer. b) Margins between the true class W_{y_i} and a topologically distant class W_{farClass} vs. a nearby class $W_{\text{nearClass}}$.

3.1 Structure-Informed Additive (Angular) Margin Losses (SIA(A)M)

Our method can be seen as an extension of Equation 2, where all classes are subject to a penalty based on their label structural distance to the true label. More precisely:

$$\Psi = \begin{cases} \cos(\theta_j) + \mu \cdot CTM_{j,y_i} & \text{if } \mathcal{L}_{\text{SIAM}} \\ \cos(\theta_j - \mu \cdot CTM_{j,y_i}) & \text{if } \mathcal{L}_{\text{SIAAM}} \end{cases} \tag{6}$$

where μ is a scalar controlling the strength of the penalty and the Class Topology Matrix CTM is a non-negative distance matrix that represents the structural label topology. During training, the true class is penalized via m, pushing its decision boundary away from the embedding. Non-target class logits are adjusted according to their distance to the true class in the CTM, as illustrated in Figure 1. Distant classes are suppressed more than nearby ones, forcing the model to account for the label structure.

A simple ordinal structure where the distance between two ordered classes is 1 can be initialized by $CTM_{j,y_i} = |j - y_i|$ yielding a distance of $CTM_{1,2} = 1$ between class $j = 1$ and true class $y_i = 2$, and $CTM_{10,3} = 7$ between class $j = 10$ and true class $y_i = 3$. A simple circular topology can be initialized using circular geodesic distance $CTM_{j,y_i} = min(|j - y_i|, C - |j - y_i|)$ and, given $C = 10$, would yield $CTM_{1,2} = 1$ and $CTM_{10,3} = 3$.

The matrix can encode more complex class relationships beyond simple integer distances. This is particularly useful in ordinal problems where the spacing between adjacent classes is not uniform. With $CTM = \mathbf{0}$, SIAM and SIAAM generalize AM-Softmax and AAM-Softmax respectively. Moreover, placing $-m$ in the diagonal of the distance matrix, such that $mCTM = -\frac{m}{\mu}\boldsymbol{I}_n + CTM$, the equations can be simplified as

$$\mathcal{L}_{\text{A(A)M}} = -\frac{1}{n}\sum_{i=1}^{n}\log\frac{e^{\,s\Psi}}{\sum_{j=1}^{C} e^{\,s\Psi}} \tag{7}$$

with

$$\Psi = \begin{cases} \cos(\theta_j) + \mu \cdot mCTM_{j,y_i} & \text{if } \mathcal{L}_{\text{SIAM}} \\ \cos(\theta_j - \mu \cdot mCTM_{j,y_i}) & \text{if } \mathcal{L}_{\text{SIAAM}} \end{cases} \tag{8}$$

4 Experimental Setup

We choose three speech classification tasks in which a structure exists in the label space. Age Detection (AD) and Lyric Intelligibility Prediction (LIP) can be formulated as ordinal classification tasks, while Speech Emotion Recognition (SER) can be formulated as a circular problem. In this section, we briefly present each task and dataset(s) used for each experiment. Model implementations for all tasks are detailed in Section 4.4.

4.1 Ordinal Problem: Age detection - AD

As a first experimental scenario, we consider a classical ordinal regression problem, namely age prediction from speech, and reformulate it as an age range classification task. For this, we aggregated several speech datasets for which the age of the speaker is provided. Table 1 summarizes the aggregated corpora. For datasets with range-based annotations (R), the original age groupings are as follows: FCS provides three groups (22-40, 40-65, 65+), PyARA uses decade-level categories (teens, twenties, ..., sixties), and Urdu 20000 provides decade-based ranges (teens to forties). Datasets marked with Y provide exact speaker ages in years. We reformulate age prediction as a four-class ordinal problem:{< 20, 20-40, 40-65, > 65}.

Nine samples from PyARA (60-70) are removed due to ambiguity between the 40-65 and > 65 categories. Unique speakers are placed in one of the train, validation or test splits whenever speaker identity is available from the original

Table 1. Age Detection datasets. Age label format (R - range, Y-years), number of unique speakers in Train/Validation/Test, and used number of utterances. Only real speech recordings were used in PyARA*.

Dataset	Age	Train	Validation	Test	Utterances
FCS [24]	R	67	12	18	30,043
TIMIT [14, 13]	Y	450	94	85	12,580
NNCES [31]	Y	33	8	9	5,000
VCTK [9]	Y	69	19	20	43,832
EmoDB [4]	Y	6	2	2	536
PyARA* [30]	R	all	–	–	73,245
Urdu 20000 [1]	R	all	–	–	16,198
EADdemo [33]	Y	15	2	3	80

datasets. Datasets without speaker identity (PyARA and Urdu 20000) are assigned entirely to the training set, to ensure that no speakers are in more than one split. The final aggregated dataset is imbalanced. We therefore use weighted sampling during training, with one epoch defined as 25,000 sampled utterances.

4.2 Ordinal Problem: Lyric Intelligibility Prediction - LIP

Lyric intelligibility prediction estimates how accurately listeners recognize words in sung audio. The target variable, *correctness*, is defined as the proportion of correctly identified words in an audio excerpt. We use the CLIP dataset introduced in the ICASSP 2026 Cadenza Challenge [32]. It provides audio excerpts annotated with word recognition scores under simulated hearing conditions (normal hearing, mild, and moderate hearing loss). The correctness score is defined as a continuous value. Following common practice in ordinal modeling, we reformulate the regression problem as ordered classification. The correctness score is discretized into 10 ordinal classes using quantitative binning. We retain the original train/validation/test split provided by the dataset authors. Unlike ASR-based approaches, we rely solely on acoustic representations to ensure language independence and robustness to musical variability.

4.3 Circular Problem: Speech Emotion Recognition - SER

Various models of emotions have been put forward by psychologists, but there is no clear consensus about which is best suited for affective computing tasks such as SER [7, 8, 12]. One of them, the circumplex model of emotion [34], uses two perpendicular linear descriptors (Valence and Arousal) to arrange emotions on a Euclidean plane around a circle. This model has the advantage of computational simplicity (two independent linear regressors) while avoiding the subjectivity of discrete emotion labelling. The plane is often partitioned into radial sectors and the regression models are then used as a classifier based on the sector in which the point lies $(V_{\text{pred}}, A_{\text{pred}})$ [23, 3, 12].

We partition the affective space into 8 angular sectors to formulate the task as circular classification and explicitly model the circular topology. The ESCorpus-PE dataset [28], containing 3749 utterances with three dimensions (Valence, Arousal, and Dominance) in Spanish with a Peruvian accent from 80 speakers, is used for our experiment. Valence (V) and Arousal (A) annotations are rescaled to $[-1, 1]$ and angle annotations are obtained using $atan2(A, V)$. Neutral extracts ($A = V = 0$) are excluded. To reduce the risk of the model memorizing speaker or recording-specific characteristics, each source is used in either the training, validation or testing split. The training set includes Audio: 1-3, 6-8, 11, and 13, the validation set Audio: 5, 10, and 14, and the test set Audio: 4, 9, and 12.

4.4 Implementation Details

Pre-trained models are used as backbone encoders. To preserve pre-trained knowledge while allowing task-specific adaptation, 80% of the parameters are frozen. A temporal statistical pooling layer [36] computing only the mean (LIP), or both mean and standard deviation (AD, SER) of an input sequence is used to aggregate the encoder's frame-level embeddings. It then provides the input f_i of the last fully-connected layer implementing the model and loss we aim to evaluate. For models using hyperparameters m and s, grid search was used.

The goal of these experiments is to test the capacity of the proposed losses for both ordinal and circular classification tasks against other methods, and not to find the best performing settings for a task. For this reason and to reduce the hyperparameter space, we set the hyperparameter μ from Equation 6 to $\mu = m$. Initial experiments with different settings of the two hyperparameters showed that this was a reasonable choice that worked across the three different tasks. A full study of their effect over the three tasks is beyond the scope of this work. For the two tasks where the audio input is speech (AD, SER), the self-supervised speech representation model WavLM Base+ [6] is used as the encoder. It shows strong performance across a wide range of downstream speech tasks, including speaker verification, and emotion recognition. In the case of LIP where the audio input is musical extracts, the state-of-the-art music understanding model MERT-v1-95M [17] is chosen as the backbone encoder.

AdamW is used as the optimizer with a weight decay of 0.01 for all models, and a Cosine Annealing learning rate scheduler [22] is added for LIP training. For AD, models are trained on 6 seconds audio clips in batch size of 16 for 20 epochs with a patience of 5, at a learning rate of 2×10^{-4}. For LIP, 10 sec. audio clips are used in batch size of 16 for 60 epochs with a patience of 20, at a learning rates of 5×10^{-5} to 10^{-6}. For SER, 6 sec. audio clips are used in batch size of 4 for 60 epochs with a patience of 10 with different learning rates for the encoder (10^{-5}) and the predictive model (10^{-3}).

Finally, to improve robustness, stochastic waveform augmentation is applied to training audio inputs including additive noise for all tasks. Additionally, random gain perturbation and time shifting was used for AD and SER; low/high-pass filtering, and simulated reverberation for AD; and time stretching, parametric equalization, and pitch shifting for LIP.

5 Experimental Results

In addition to traditional evaluation metrics (Accuracy, F1, Precision), regression metrics (Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Normalized Correlation Coefficient (NCC)) are reported for the LIP models. To evaluate the different losses' capacities to reduce label topology errors, Ordinal Error (OE) defined as the mean absolute error between predicted $\hat{y}$ and true ordered class y defined as: $OE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i|$ is provided for the two ordinal tasks, while Circular Error (CirE) measuring the shortest circular geodesic distance on a periodic label space with C classes defined as: $CirE = \frac{1}{n}\sum_{i=1}^{n}\min\left(|\hat{y}_i - y_i|\,,\ C - |\hat{y}_i - y_i|\right)$ is reported for the circular problem.

5.1 Age Detection - AD

Table 2 shows that while not outperforming other ordinal-informed losses (CORN and CORAL) on accuracy or F1, our proposed methods achieve comparable results while outperforming all un-informed losses (cross-entropy (CE), AM, and AAM). Moreover, SIAM obtains the smallest ordinal error and highest precision. This indicates that the proposed approach, while remaining within a margin-based formulation, performs comparably to specialized ordinal methods.

Table 2. Age Detection (AD) Results. Where m and s are the margin-based hyperparameters, Accuracy (Acc), Precision (P), and Ordinal Error (OE).

Loss	s	m	Acc ↑	F1 ↑	P ↑	OE ↓
CE	-	-	72.60	47.14	51.00	0.295
CORN	-	-	74.99	**51.24**	54.26	0.265
CORAL	-	-	**75.19**	51.09	53.88	0.258
AM	30	0.20	71.11	46.96	54.52	0.341
AAM	30	0.20	73.68	50.14	52.42	0.286
SIAM	20	0.3	74.82	47.07	**57.64**	**0.257**
SIAAM	10	0.1	74.23	50.05	52.84	0.267

5.2 Lyric Intelligibility Prediction - LIP

Because the correctness score is mapped into 10 ordinal classes, class predictions needs to be mapped back to a numerical value to be evaluated with RMSE, MAE and NCC. Predicted classes are mapped to the mean correctness score for each class estimated from the training set and these values are evaluated against the original continuous targets. Table 3 shows both the best and second-best score. From this we see that CORAL and SIAM obtain the best results on all distance-aware (regression) metrics. The highest accuracy is obtained with standard cross-entropy loss, which does not explicitly model label structure. However, the low precision and higher OE might indicate poor class generalization. These results indicate that the proposed losses are comparable to specialized ordinal methods (CORAL and CORN), while keeping a margin-based formulation.

Table 3. Lyric Intelligibility Prediction (LIP): Best result is bold, the second-best result is underlined. Where m and s are the margin-based hyper-parameters, Accuracy (Acc), Precision (P), (Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Normalized Correlation Coefficient (NCC), and Ordinal Error (OE).

Loss	s	m	Acc ↑	P ↑	F1 ↑	RMSE ↓	MAE ↓	NCC ↑	OE ↓
CE	-	-	**29.77**	11.05	16.09	0.490	0.360	0.363	3.207
CORAL	-	-	14.70	<u>19.92</u>	14.53	**0.334**	**0.265**	**0.453**	**2.372**
CORN	-	-	17.35	19.56	<u>17.61</u>	0.352	0.273	0.431	2.447
SIAM	40	0.10	17.08	19.30	16.97	<u>0.335</u>	<u>0.266</u>	<u>0.447</u>	<u>2.384</u>
SIAM	10	0.20	15.89	**22.19**	16.06	0.357	0.281	0.395	2.534
SIAAM	20	0.02	<u>22.28</u>	18.47	**19.72**	0.390	0.295	0.406	2.642

5.3 Speech Emotion Recognition - SER

Since CORAL and CORN are designed for ordinal settings, we compare our methods against cross-entropy and a Concordance Correlation Coefficient (CCC) loss-based regression typically used in SER [39, 40, 2]. Table 4 shows our methods outperform the baselines under identical architectures and training. Moreover, they improve both AM-Softmax and AAM-Softmax performances consistently across classification metrics and circular error (CirE), indicating that explicit modeling of circular topology leads to better alignment with the label structure.

Table 4. Speech Emotion Recognition (SER) Results. Where m and s are the margin-based hyper-parameters, Accuracy (Acc), Precision (P), Circular Error (CirE)

Loss	s	m	Acc ↑	F1 ↑	P ↑	CirE ↓
CE	-	-	20.68	11.15	12.42	1.478
CCC	20	-	19.14	13.57	18.65	1.423
AM	40	0.20	16.98	10.74	13.16	1.574
AAM	30	0.20	17.59	11.43	13.21	1.559
SIAM	20	0.20	20.06	20.34	34.27	1.327
SIAM	30	0.30	19.75	**20.84**	**34.90**	1.324
SIAAM	30	0.05	19.75	14.17	17.68	**1.312**
SIAAM	30	0.10	**22.53**	17.02	18.03	1.349

6 Conclusions and Future Work

We generalized the popular AM-Softmax and AAM-softmax losses by introducing a Class Topology Matrix that can encode complex class relationship including ordinal and circular. Unlike conventional angular margin losses, which enforce uniform inter-class separation, SIA(A)M introduces a topology-aware constraint

that reflects structural relationships among classes. This enables the learned representation to preserve meaningful relational structure in addition to discriminative power and enforce non-uniform inter-class relationships. The CTM can conveniently be integrated in existing margin-based loss, and SIAM and SIAAM reduce to AM-Softmax and AAM-Softmax when CTM = 0. Across two ordinal classification tasks, the proposed methods match the performance and structural error (OE) of specialized ordinal loss functions. On a circular classification task, for which no direct counterpart to CORAL or CORN currently exists, our methods outperform all standard classification baselines.

In the future, we plan to investigate additional ordinal and circular tasks as well as more complex relationship structures such as graphs and hierarchies. A detailed experimental investigation of different hyper-parameter settings will help derive guidelines for different tasks. A more thorough comparison with regression approaches followed by discretizations of the output is also something we would like to investigate. Another possible interesting direction for future work is to explore learning the class topology matrices (CTM) from the data. A learned CTM might be slightly different than the actual problem CTM which leads to the question of how sensitive is the resulting trained model to the accuracy of the CTM. We plan to conduct sensitivity experiments to explore this question. Finally, there is an additional computational cost when introducing the CTM for the calculation of the loss function and it can potentially lead to slower convergence (more epochs). We plan to conduct a detailed theoretical and experimental analysis of the computational cost in the future.

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